



Caringbah High School

Year 12 2024

Mathematics Extension 1

HSC Course

Assessment Task 2

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using a black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 6-10, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 45

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 40 marks

Attempt Questions 6-10
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II					Total	
Q 1-5	Q6	Q7	Q8	Q9	Q10		
							%
/5	/8	/9	/8	/7	/8	/45	

Section I

5 marks

Attempt Questions 1-5

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1 Which of the following vectors is parallel to $\underline{u} = \underline{i} - 2\underline{j}$?

- (A) $\underline{i} + 2\underline{j}$
- (B) $-\underline{i} - 2\underline{j}$
- (C) $-3\underline{i} + 6\underline{j}$
- (D) $5\underline{i} + 10\underline{j}$

2 Find $|\underline{y}|$ for the vector $\underline{y} = -\underline{i} - 6\underline{j}$.

- (A) 7
- (B) $\sqrt{7}$
- (C) 37
- (D) $\sqrt{37}$

3 Which of the following vectors has a magnitude of 5?

- (A) $\frac{\sqrt{2}}{10}(-5\underline{i} + 5\underline{j})$
- (B) $5(-5\underline{i} + 5\underline{j})$
- (C) $\frac{\sqrt{2}}{2}(-5\underline{i} + 5\underline{j})$
- (D) $\frac{5\sqrt{2}}{2}(-5\underline{i} + 5\underline{j})$

- 4 The vectors $\underline{a} = x\underline{i} - 2\underline{j}$ and $\underline{b} = -6\underline{i} + y\underline{j}$ are perpendicular.
What are the possible values of x and y ?
- (A) $x = 1$ and $y = 3$
(B) $x = 1$ and $y = -3$
(C) $x = -2$ and $y = -6$
(D) $x = 2$ and $y = 6$
- 5 A ship leaves a port O and sails on a bearing of $N30^\circ E$ for 100 km to a point P .
If $\underline{i}$ and $\underline{j}$ are unit vectors in the east and north directions respectively, the
position vector $\overrightarrow{OP}$ is equal to:
- (A) $50\sqrt{3}\underline{i} + 50\underline{j}$
(B) $70\underline{i} + 70\underline{j}$
(C) $30\underline{i} + 70\underline{j}$
(D) $50\underline{i} + 50\sqrt{3}\underline{j}$

End of Section I

Section II

40 marks

Attempt Questions 6–10

Allow about 65 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (8 marks) Use a SEPARATE writing booklet.

(a) If $\underline{u} = 3\underline{i} - \underline{j}$ and $\underline{v} = \underline{i} - 2\underline{j}$, find:

(i) $\underline{u} - \underline{v}$ 1

(ii) $(\underline{u} - \underline{v}) \cdot \underline{v}$ 1

(iii) Hence, what can be said about the vectors $\underline{v}$ and $\underline{u} - \underline{v}$? 1

(b) The position vectors of points A and B are $\overrightarrow{OA} = 8\underline{i} - 3\underline{j}$ and $\overrightarrow{OB} = -5\underline{i} - 6\underline{j}$.

(i) Find $\overrightarrow{AB}$ 1

(ii) Hence find $|\overrightarrow{AB} - \overrightarrow{OA}|$ 2

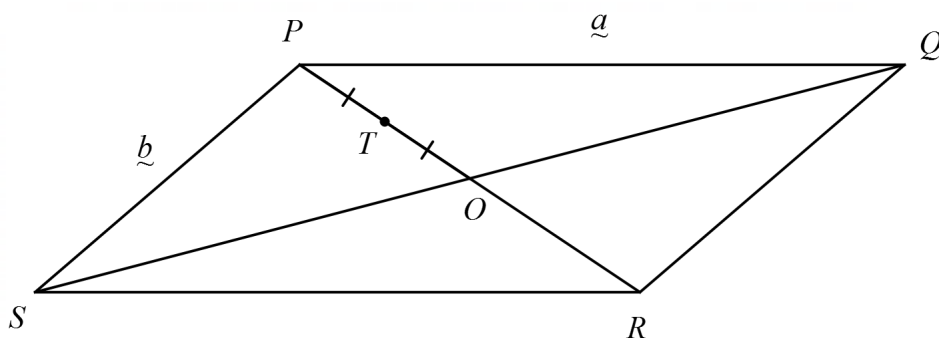
(c) Find the unit vector $\hat{v}$ in the direction of $\underline{v} = 3\underline{i} - 4\underline{j}$. 2

End of Question 6

Question 7 (9 marks) Use a SEPARATE writing booklet.

- (a) By substituting $u = 1 + x^2$, find $\int 6x\sqrt{1+x^2} dx$. 2

- (b) $PQRS$ is a parallelogram with $\overrightarrow{PQ} = \underline{a}$ and $\overrightarrow{PS} = \underline{b}$. The diagonals PR and SQ intersect at the point O . The point T is the midpoint of OP .



Find $\overrightarrow{ST}$ in terms of $\underline{a}$ and $\underline{b}$. 2

- (c) By substituting $u = x + e$, find $\int \frac{x}{(x+e)^2} dx$ 2

- (d) The vectors $\underline{u}$ and $\underline{v}$ are such that $|\underline{u}| = 3$ and $|\underline{v}| = 7$. 3

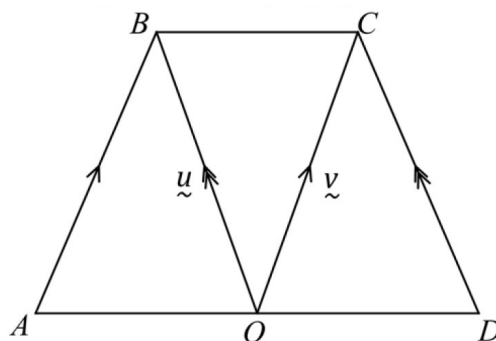
If $\underline{u} \cdot (\underline{u} + \underline{v}) = 25$, determine the size of the acute angle between the vectors $\underline{u}$ and $\underline{v}$, correct to the nearest degree.

End of Question 7

Question 8 (8 marks) Use a SEPARATE writing booklet.

- (a) In the following diagram $\overline{AB}$ is parallel to $\overline{OC}$ and $\overline{DC}$ is parallel to $\overline{OB}$.

$$\overline{OB} = \underline{u}, \quad \overline{OC} = \underline{v} \quad \text{and} \quad |\overline{AB}| = |\overline{OB}| = |\overline{OC}| = |\overline{DC}|.$$



Express $\overline{DB}$ in terms of $\underline{u}$ and $\underline{v}$.

2

- (b) For $\underline{a} = -3\underline{i} - 2\underline{j}$ and $\underline{b} = 4\underline{i} + 5\underline{j}$, find:

(i) $\underline{a} \cdot \underline{b}$

1

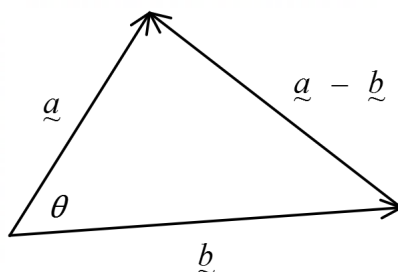
- (ii) the vector projection of $\underline{a}$ onto $\underline{b}$

2

- (c) In the following triangle, use vectors to prove the Cosine Rule,

3

$$|\underline{a} - \underline{b}|^2 = |\underline{a}|^2 + |\underline{b}|^2 - 2|\underline{a}||\underline{b}|\cos\theta$$



End of Question 8

Question 9 (7 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_1^{\sqrt{5}} \frac{x}{\sqrt{1+3x^2}} dx$ using the substitution $u = 1 + 3x^2$ **3**

(b) Consider the points $P(a, 2a)$, $Q(-a, 5a)$, $R(3a, 4a)$ and $S(9a, 12a)$, where a is a positive real number.

(i) Express $\overrightarrow{PQ}$ in component form. **1**

(ii) Given the length of the projection of $\overrightarrow{PQ}$ onto $\overrightarrow{RS}$ is 6, find the value of a . **3**

End of Question 9

Question 10 (8 marks) Use a SEPARATE writing booklet.

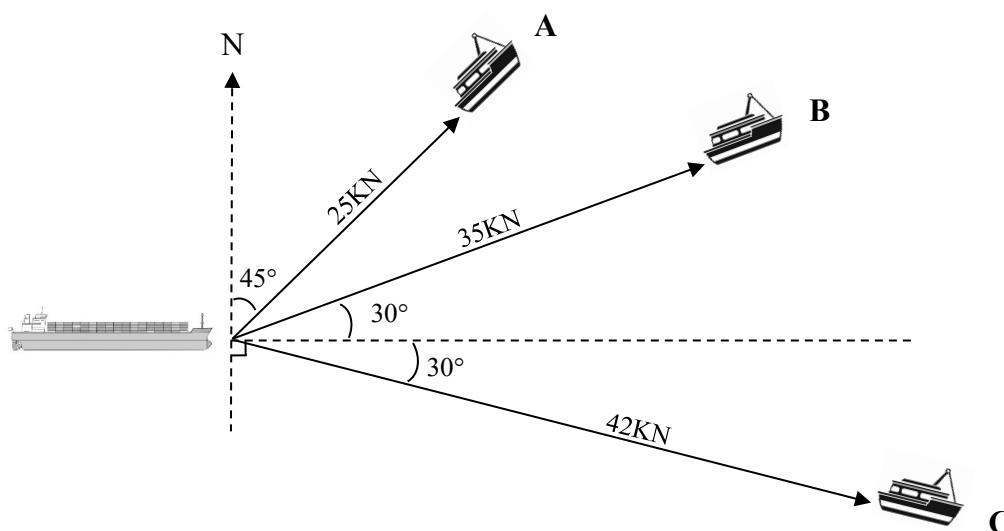
(a) Evaluate $\int_3^8 \frac{x}{\sqrt{1+x}} dx$ using the substitution $x = u - 1$ 3

- (b) Three tugboats are pulling a much larger container ship out of port as represented in the diagram below.

Tugboat A is pulling at a force of 25KN,

Tugboat B is pulling at a force of 35KN and

Tugboat C is pulling at a force of 42KN.



- (i) Show that the resultant force can be represented by

$$\vec{F} = \frac{1}{2} \left[(25\sqrt{2} + 77\sqrt{3})\vec{i} + (25\sqrt{2} - 7)\vec{j} \right] \quad 3$$

- (ii) Calculate the magnitude and direction of $\vec{F}$, correct to the nearest KN and degree respectively. 2

End of Exam

Name: _____

Mathematics Extension 1

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct ↗

-
- | | | | | | |
|--------------------|----|-------------------------|-------------------------|-------------------------|-------------------------|
| Start
→
Here | 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| | 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| | 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| | 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| | 5. | A ☐ | B ☐ | C ☐ | D ☐ |

2024 Year 12 Extension 1 AT2 Solutions

Section I

Q1) C

Q2) D

$$|v| = \sqrt{(-1)^2 + 6^2}$$

$$= \sqrt{37}$$

Q3) C

$$\frac{\sqrt{2}}{2} \times \sqrt{(-5)^2 + 5^2}$$

$$= \frac{\sqrt{2}}{2} \times 5\sqrt{2}$$

$$= 5$$

Q4) B

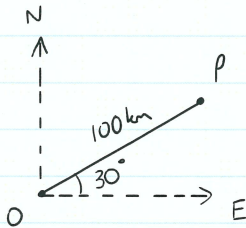
$$\underline{a} \cdot \underline{b} = 0$$

$$-6x - 2y = 0$$

$$6x = -2y$$

$$\therefore x = 1 \text{ and } y = -3$$

Q5) A



$$\sqrt{(50\sqrt{3})^2 + 50^2}$$

$$= 100$$

$\vec{OP} = 50\sqrt{3}\underline{i} + 50\underline{j}$ since the horizontal component is clearly longer than the vertical component in our case.

Section II

$$\begin{aligned} \text{Q6) a) i) } \underline{u} - \underline{v} &= 3\underline{i} - \underline{j} - (\underline{i} - 2\underline{j}) \\ &= 3\underline{i} - \underline{j} - \underline{i} + 2\underline{j} \\ &= 2\underline{i} + \underline{j} \end{aligned}$$

$$\begin{aligned} \text{ii) } (\underline{u} - \underline{v}) \cdot \underline{v} &= (2\underline{i} + \underline{j}) \cdot (\underline{i} - 2\underline{j}) \\ &= 2 \times 1 + 1 \times -2 \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

iii) They are perpendicular.

$$\begin{aligned} \text{b) i) } \overrightarrow{OA} &= 8\underline{i} - 3\underline{j} \\ \overrightarrow{OB} &= -5\underline{i} - 6\underline{j} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AB} &= (-5-8)\underline{i} + (-6--3)\underline{j} \\ &= -13\underline{i} - 3\underline{j} \end{aligned}$$

$$\begin{aligned} \text{ii) } |\overrightarrow{AB} - \overrightarrow{OA}| & \\ &= |-13\underline{i} - 3\underline{j} - (8\underline{i} - 3\underline{j})| \\ &= |-13\underline{i} - 3\underline{j} - 8\underline{i} + 3\underline{j}| \\ &= |-21\underline{i}| & \textcircled{1} \\ &= \sqrt{21^2} \\ &= 21 & \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{c) } |\underline{v}| &= \sqrt{3^2 + (-4)^2} \\ &= 5 & \textcircled{1} \end{aligned}$$

$$\hat{\underline{v}} = \frac{1}{5} \times (3\underline{i} - 4\underline{j}) \quad \textcircled{1}$$

$$\hat{\underline{v}} = \frac{3}{5}\underline{i} - \frac{4}{5}\underline{j}$$

$$Q7) a) u = 1 + x^2 \quad x^2 = u - 1$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$3 \int u^{\frac{1}{2}} du$$

① Substituting

$$= 3 \times \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= 2u^{\frac{3}{2}} + C$$

$$= 2(1+x^2)^{\frac{3}{2}} + C$$

①

$$b) \vec{ST} = \vec{SP} + \vec{PT}$$

$$= -\underline{\underline{b}} + \vec{PT}$$

$$\vec{PT} = \frac{1}{4}(\vec{PS} + \vec{SR})$$

$$= \frac{1}{4}(\underline{\underline{b}} + \underline{\underline{a}}) \quad \text{①}$$

$$\therefore \vec{ST} = -\underline{\underline{b}} + \frac{1}{4}(\underline{\underline{b}} + \underline{\underline{a}})$$

$$= \frac{1}{4}\underline{\underline{a}} - \frac{3}{4}\underline{\underline{b}} \quad \text{①}$$

$$c) u = x + e \quad x = u - e$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$\int \frac{u-e}{u^2} du$$

①

$$\int \frac{1}{u} - e u^{-2} du$$

$$= \ln u + \frac{e}{u} + C$$

$$= \ln|x+e| + \frac{e}{x+e} + C$$

①

$$d) \quad \underline{u} \cdot (\underline{u} + \underline{v}) = 25$$

$$\underline{u} \cdot \underline{u} + \underline{u} \cdot \underline{v} = 25$$

$$\underline{u} \cdot \underline{v} = 25 - |\underline{u}|^2$$

$$\underline{u} \cdot \underline{v} = 25 - 3^2$$

$$\underline{u} \cdot \underline{v} = 16$$

$$\text{since } \underline{u} \cdot \underline{u} = |\underline{u}|^2 \quad (1)$$

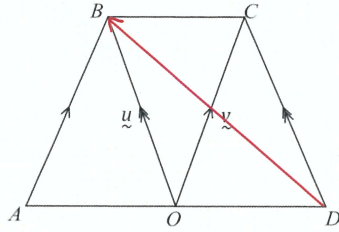
$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\therefore \cos \theta = \frac{16}{3 \times 7}$$

$$\theta = \cos^{-1} \left(\frac{16}{21} \right)$$

$$\theta = 40^\circ \quad (1)$$

Q8) a)



$$\begin{aligned}\vec{OB} &= \vec{OO} + \vec{OB} \\ &= \vec{u} - \vec{v} + \vec{u} \\ &= 2\vec{u} - \vec{v}\end{aligned}\quad (1)$$

$$\begin{aligned}\vec{OD} &= \vec{OC} + \vec{CO} \\ &= \vec{u} - \vec{v}\end{aligned}\quad (1)$$

b) i) $\vec{a} = -3\hat{i} - 2\hat{j}$
 $\vec{b} = 4\hat{i} + 5\hat{j}$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= -3 \times 4 + -2 \times 5 \\ &= -12 - 10 \\ &= -22\end{aligned}$$

ii) $\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} \vec{b}$

$$= \frac{-22}{4^2 + 5^2} (4\hat{i} + 5\hat{j}) \quad (1)$$

$$= \frac{-22}{41} (4\hat{i} + 5\hat{j}) \quad (1)$$

c) LHS = $|\vec{a} - \vec{b}|^2$

$$\begin{aligned}&= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) \quad (1) \\ &= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} \quad (1) \\ &= |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos\theta \quad (1) \\ &= \text{RHS}\end{aligned}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

$$\begin{aligned} \text{Q9) a) } u &= 1 + 3x^2 \\ \frac{du}{dx} &= 6x \\ x dx &= \frac{du}{6} \end{aligned}$$

$$\begin{aligned} u &= 1 + 3 \times 1^2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} u &= 1 + 3 \times (\sqrt{5})^2 \\ &= 16 \end{aligned}$$

$$\frac{1}{6} \int_4^{16} u^{-\frac{1}{2}} du \quad \textcircled{1}$$

$$= \frac{1}{6} \left[2u^{\frac{1}{2}} \right]_4^{16} \quad \textcircled{1}$$

$$= \frac{1}{3} (\sqrt{16} - \sqrt{4})$$

$$= \frac{1}{3} \times (4 - 2)$$

$$= \frac{2}{3} \quad \textcircled{1}$$

$$\begin{aligned} \text{b) i)} \quad \vec{PQ} &= (-a-a)\underline{i} + (5a-2a)\underline{j} \\ &= -2a\underline{i} + 3a\underline{j} \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad \vec{PQ} &= -2a\underline{i} + 3a\underline{j} \\ \vec{RS} &= 6a\underline{i} + 8a\underline{j} \end{aligned}$$

$$\begin{aligned} \vec{PQ} \cdot \vec{RS} &= -2a \times 6a + 3a \times 8a \\ &= -12a^2 + 24a^2 \\ &= 12a^2 \end{aligned} \quad (1)$$

$$\begin{aligned} |\vec{RS}| &= \sqrt{(6a)^2 + (8a)^2} \\ &= \sqrt{100a^2} \\ &= 10a \end{aligned} \quad (1)$$

Method 1

$$\text{Proj}_{\vec{RS}} \vec{PQ} = \frac{\vec{PQ} \cdot \vec{RS}}{|\vec{RS}|}$$

$$6 = \frac{12a^2}{10a}$$

$$\frac{6}{5}a = 6$$

$$a = 5 \quad (1)$$

Method 2

$$\text{Proj}_{\vec{RS}} \vec{PQ} = \frac{\vec{PQ} \cdot \vec{RS}}{\vec{RS} \cdot \vec{RS}} \times \vec{RS}$$

$$\underline{v} = \frac{12a^2}{100a^2} (6a\underline{i} + 8a\underline{j})$$

$$= \frac{3}{25} (6a\underline{i} + 8a\underline{j})$$

$$|\underline{v}| = 6 = \frac{3}{25} \times \sqrt{(6a)^2 + (8a)^2}$$

$$6 = \frac{3}{25} \times 10a$$

$$10a = 50$$

$$a = 5$$

$$\text{Q10) a) } x = u - 1 \\ u = x + 1$$

$$u = x + 1 \\ \frac{du}{dx} = 1 \\ du = dx$$

$$u = x + 1 \\ u = 5 + 1 \\ = 6$$

$$u = x + 1 \\ u = 10 + 1 \\ = 11$$

$$\int_4^9 \frac{u-1}{\sqrt{u}} du \quad (1)$$

$$\int_4^9 \frac{u}{u^{\frac{1}{2}}} - \frac{1}{u^{\frac{1}{2}}} du$$

$$\int_4^9 u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= \left[\frac{2u^{\frac{3}{2}}}{3} - 2u^{\frac{1}{2}} \right] \quad (1)$$

$$= 2 \left[\frac{u^{\frac{3}{2}}}{3} - u^{\frac{1}{2}} \right]_4^9$$

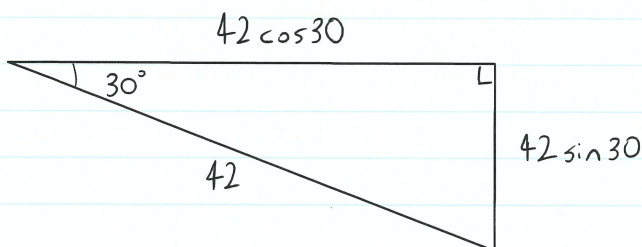
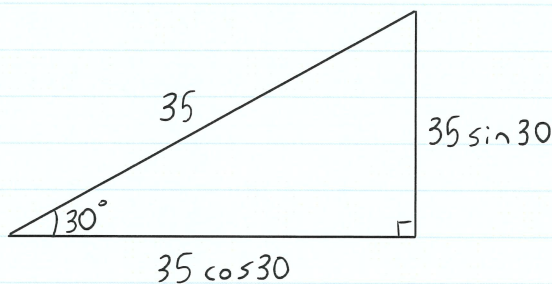
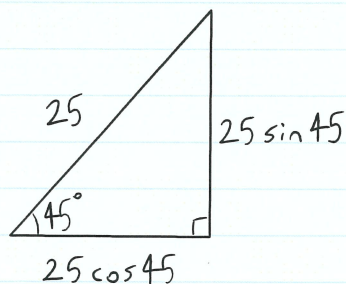
$$= 2 \left[\left(\frac{9^{\frac{3}{2}}}{3} - \sqrt{9} \right) - \left(\frac{4^{\frac{3}{2}}}{3} - \sqrt{4} \right) \right]$$

$$= 2 \left[\left(\frac{27}{3} - 3 \right) - \left(\frac{8}{3} - 2 \right) \right]$$

$$= 2 \left(6 - \frac{2}{3} \right)$$

$$= \frac{32}{3} \quad (1)$$

b) i)



① for calculating horizontal and vertical components

$$\begin{aligned}\vec{F} &= (25 \cos 45 + 35 \cos 30 + 42 \cos 30) \hat{i} + (25 \sin 45 + 35 \sin 30 - 42 \sin 30) \hat{j} \quad \text{①} \\ &= \left(25 \times \frac{1}{\sqrt{2}} + 35 \times \frac{\sqrt{3}}{2} + 42 \times \frac{\sqrt{3}}{2} \right) \hat{i} + \left(25 \times \frac{1}{\sqrt{2}} + 35 \times \frac{1}{2} - 42 \times \frac{1}{2} \right) \hat{j} \\ &= \left(\frac{25\sqrt{2}}{2} + \frac{77\sqrt{3}}{2} \right) \hat{i} + \left(\frac{25\sqrt{2}}{2} - \frac{7}{2} \right) \hat{j} \\ &= \frac{1}{2} \left[(25\sqrt{2} + 77\sqrt{3}) \hat{i} + (25\sqrt{2} - 7) \hat{j} \right] \quad \text{as required.} \quad \text{①}\end{aligned}$$

$$\begin{aligned}\text{ii) } |\vec{F}| &= \frac{1}{2} \times \sqrt{(25\sqrt{2} + 77\sqrt{3})^2 + (25\sqrt{2} - 7)^2} \\ &= 85.55 \text{ KN}\end{aligned}$$

① Accept answers rounded to the nearest KN.

$$\theta = \tan^{-1} \left(\frac{25\sqrt{2} - 7}{25\sqrt{2} + 77\sqrt{3}} \right)$$

$$= \text{N } 9^{\circ} 32' \text{ E}$$

① Accept answers rounded to the nearest degree.

Also accept $080^{\circ} 28' \text{ T}$